

Gate composer

Boolean functions become gate networks, AND, OR, NOT, and friends wired together

Wire, probe, check

- Place gates, connect inputs A through D, and read output F on every row
- NAND is NOT of AND; XOR is one when inputs differ
- A two-to-one mux selects B or C under A
- Full-adder sum is $A \text{ xor } B \text{ xor } C_{in}$; carry-out is majority
- The truth table is the judge, if every row matches, the net is correct

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Gate composer

Build a small combinational net from AND, OR, XOR, and NOT. The netlist is compiled to continuous assigns by the HDL engine. Twenty-nine challenges from AND through 2-bit equality and decode.

Starter example: a single AND ($F = A \& B$). Click a challenge card to build from scratch.

Load starter example

Challenges0 / 29 cleared

INTRO

AND (2)
2 in · Compose $F = A \text{ AND } B$.

OR (2)
2 in · Compose $F = A \text{ OR } B$.

Inverter
2 in · Ignore B: set $F = \text{NOT } A$ (F follows $\sim A$ for every row).

NAND (2)
2 in · Compose $F = A \text{ NAND } B$ (NOT of AND).

Implication $A \rightarrow B$
2 in · $F = 1$ except when $A=1$ and $B=0$ (material implication).

XOR (2)
2 in · Build $A \oplus B$ ($F=1$ when A and B differ).

Equality / XNOR
2 in · $F = 1$ when A equals B.

NOR (2)
2 in · Compose $F = A \text{ NOR } B$.

CORE

Exactly one (2)
2 in · Build $F = A \oplus B$ (F is 1 when exactly one of A and B is 1).

Majority (3)
3 in · Build $F = A \& B \vee A \& C \vee B \& C$ (F is 1 when at least two of A, B, C are 1).

Exactly one (3)
3 in · Build $F = A \oplus B \oplus C$ (F is 1 when an odd number of inputs are 1).

Gate composer starter

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Workbook practice

- In the workbook track, sketch a two-input XOR from AND, OR
- Draw a three-input majority gate net
- Write the half-adder sum and carry as gate names
- Name one pitfall: a net that works on three rows but fails on the fourth

Pitfalls to watch

- Do not leave inputs floating, every gate input must be tied
- Reusing the wrong polarity breaks active-high logic
- And remember: the browser lab is literacy
- Real RTL still needs synthesis, timing, and test vectors beyond one truth table

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one small gate net and label wires
- When you are ready, take the short quiz, then continue to mux and decoder